

Quiz3, MAT1575 Professor Chiu

ID: _____

Name: Sol

- This quiz consists of 5 questions for a total of 10 points.
- Show all work and justify your answers. You have 15 minutes to complete the quiz.
- Wishing you success.

(10 points) Evaluate each of the following integrals:

$$(1) \int_{-1}^2 (x^2 + 3x) dx = \left[\frac{x^3}{3} + \frac{3}{2}x^2 \right]_{-1}^2 = \left[\frac{(2)^3}{3} + \frac{3}{2}(2)^2 \right] - \left[\frac{(-1)^3}{3} + \frac{3}{2}(-1)^2 \right]$$

$$= \left[\frac{8}{3} + 6 \right] - \left[-\frac{1}{3} + \frac{3}{2} \right] = \frac{8}{3} + 6 + \frac{1}{3} - \frac{3}{2} = \frac{8}{3} + \frac{1}{3} + 6 - \frac{3}{2} = 9 - \frac{3}{2} = \frac{15}{2}$$

$$(2) \int_0^{\frac{\pi}{4}} \sec \theta \tan \theta d\theta = \left[\sec \theta \right]_0^{\frac{\pi}{4}} = \sec\left(\frac{\pi}{4}\right) - \sec(0) = \frac{1}{\cos(\frac{\pi}{4})} - \frac{1}{\cos(0)}$$

$$= \frac{2}{\sqrt{2}} - 1 = \sqrt{2} - 1$$

$$(3) \int (x-1)^5 dx = \int u^5 du = \frac{u^6}{6} + C = \frac{(x-1)^6}{6} + C$$

$u = x-1$
 $du = dx$

$$(4) \int \frac{x}{\sqrt{x^2+1}} dx = \int \frac{1}{\sqrt{u}} \frac{du}{2} = \frac{1}{2} \int u^{-\frac{1}{2}} du = \frac{1}{2} \cdot 2u^{\frac{1}{2}} + C$$

$$u = x^2+1$$

$$du = 2x dx$$

$$\frac{du}{2} = x dx$$

$$= u^{\frac{1}{2}} + C = (x^2+1)^{\frac{1}{2}} + C$$

$$(5) \int_0^{\frac{\pi}{4}} \frac{\cos \theta}{\sin^4 \theta} d\theta = \int_{u(0)}^{u(\frac{\pi}{4})} \frac{du}{u^4} = \int_0^{\frac{\sqrt{2}}{2}} u^{-4} du = \frac{u^{-3}}{-3} \Big|_0^{\frac{\sqrt{2}}{2}} = -\frac{1}{3u^3} \Big|_0^{\frac{\sqrt{2}}{2}}$$

$u = \sin \theta$
 $du = \cos \theta d\theta$

$$u\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$u(0) = \sin(0) = 0$$

$$= \left[-\frac{1}{3\left(\frac{\sqrt{2}}{2}\right)^3} - \lim_{u \rightarrow 0} \left[-\frac{1}{3u^3} \right] \right] \Rightarrow \text{DO NOT EXIST.}$$

DO NOT EXIST

End of this quiz.